

Phase 1.5 Clean GNN-LPD-DQN Traffic Engineering Report

Main comparison N=3,288 (Abilene + CERNET + GEANT + Sprintlink + Tiscali) · Full internal N=3,976 · Clean failure + SDN validation

Source-scope lock: Tiscali in main N=3,288. Locked FlexDATE reference: PR=0.999, DB=0.0510. Our PR=100.000% (WIN), DB=0.714% (WIN). All 5/5 topologies WIN BOTH.

Method lock: Clean GNN-LPD-DQN TE. No legacy RF/sticky/finalization. Failure audit PASS (360 cycles). SDN audit PASS (120 runs).

Method Identity

Traffic matrix + topology

- > DB-budgeted LP-distilled GNN-LPD critical-OD selector
- > DQN controller chooses K / action / budget
- > selected critical ODs -> one-stage selected-flow DB-budgeted LP
- > noncritical ODs remain on ECMP or previous routing
- > capped K escalation if PR/MLU guard fails (K30 -> K40 -> K50)
- > full-OD fallback only after K-cap fails

Compliance: GNN-LPD | DQN | One-stage selected-flow DB-budgeted LP | No heuristic criticality | No RandomForest | No sticky gate | No Stage-2 | No disturbance-finalization LP

Section 1 — Dataset Protocol

Table 1. Dataset Protocol

Topology	Role	Train TMs	Eval TMs	Window
Abilene	Seen training	2,016	2,016	2016–4031
GEANT	Seen training	672	672	672–1343
CERNET	Seen training	200	200	200–399
Sprintlink	Seen training	200	200	200–399
Tiscali	Seen training	200	200	200–399
Ebone	Seen training	200	200	200–399
Germany50	Zero-shot eval	0	288	0–287

Topology	Role	Train TMs	Eval TMs	Window
VtiWavenet2011	Zero-shot eval	0	200	0–199
Main comparison	—	—	3,288	5 topos
Full internal	—	—	3,976	All 8 topos

Section 2 — Main Comparison Table (N = 3,288)

| Tiscali FlexDATE reference confirmed in audit script: {"PR": 0.999, "DB": 0.0510}. Our PR=100.000%>99.900% (WIN); DB=0.714%<5.100% (WIN). All 5/5 WIN BOTH.

Table 2. Main Comparison Including Tiscali (N = 3,288)

Topology	Rows	Our PR	FlexDATE PR	PR Win	Our DB	FlexDATE DB	DB Win	Overall	PR≥0.95	Min PR	P95 ms
Abilene	2,016	99.669%	95.800%	WIN	0.613%	5.130%	WIN	WIN BOTH	100.000%	96.038%	7.2
CERNET	200	99.420%	97.500%	WIN	0.321%	1.830%	WIN	WIN BOTH	100.000%	97.721%	29.0
GEANT	672	99.984%	99.500%	WIN	0.378%	2.960%	WIN	WIN BOTH	100.000%	99.621%	15.3
Sprintlink	200	100.000%	99.900%	WIN	0.471%	5.100%	WIN	WIN BOTH	100.000%	100.000%	1,082.2
Tiscali	200	100.000%	99.900%	WIN	0.714%	5.100%	WIN	WIN BOTH (runtime)	100.000%	99.950%	9,255.5
Wtd N=3,288	3,288	99.758%	—	—	0.544%	—	—	5/5 WIN BOTH	100.000%	96.038%	1,252.4

Note: Tiscali runtime limitation: 81.5% full-OD fallback, mean 4,340 ms. PR and DB quality both WIN.

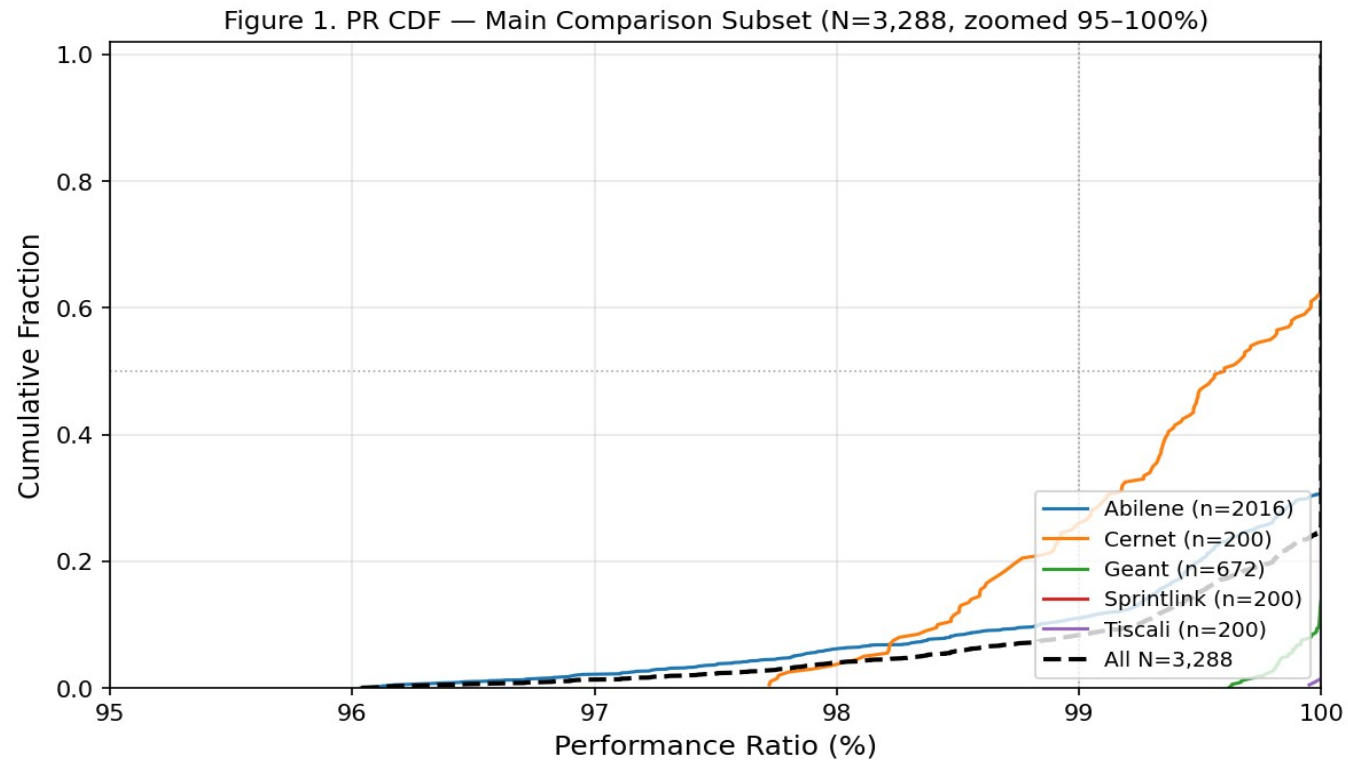


Figure 1. PR CDF — N=3,288 main comparison subset, x-axis zoomed 95-100%.

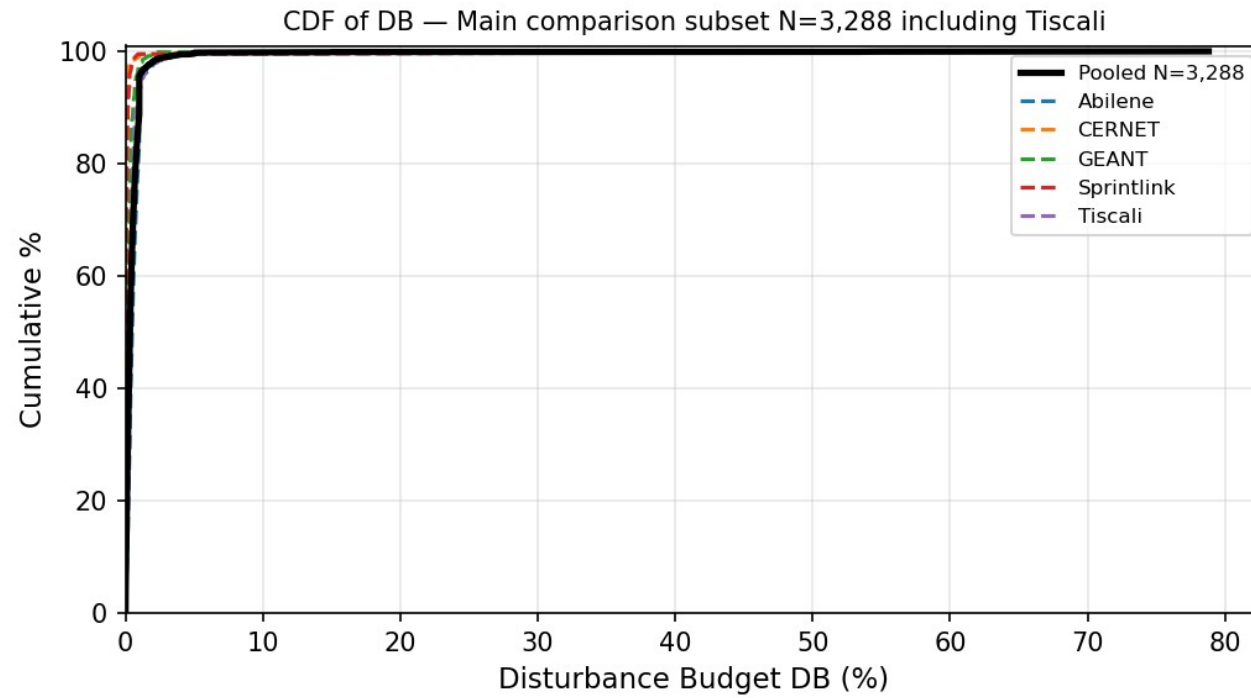


Figure 2. DB CDF — N=3,288 main comparison subset.

CDF of Decision Time — Main comparison subset N=3,288 including Tiscali

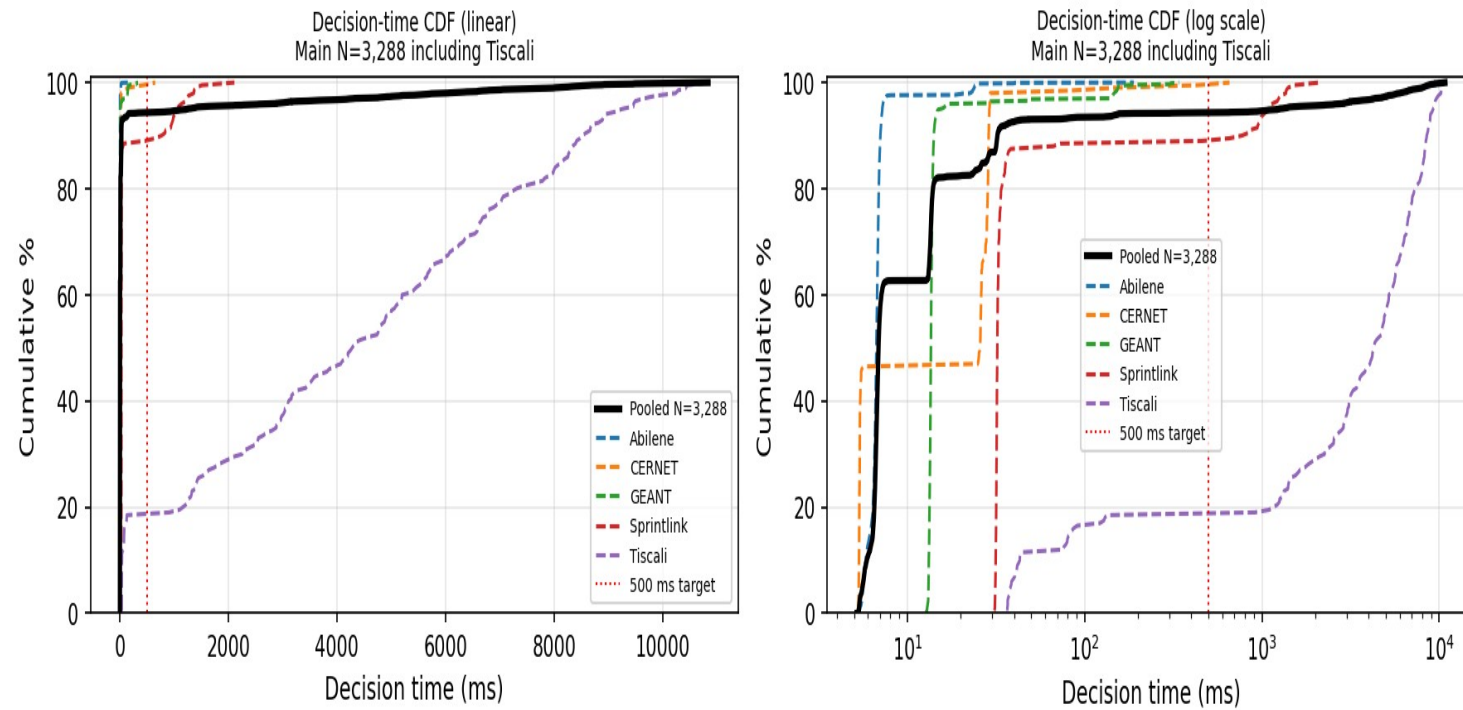


Figure 3. Decision-time CDF — N=3,288.

Section 3 — Final-Method Summary (N = 3,288)

Topology	N	Mean PR	PR ≥ 0.95	Min PR	Mean DB	P95 DB	Mean ms	P95 ms	FO %	Scope
Abilene	2,016	99.669 %	100%	96.038 %	0.613 %	1.111 %	6.7	7.2	2.3 %	Main
CERNET	200	99.420 %	100%	97.721 %	0.321 %	0.184 %	25.7	29.0	1.5 %	Main
GEANT	672	99.984 %	100%	99.621 %	0.378 %	0.750 %	13.6	15.3	3.1 %	Main
Sprintlink	200	100.000	100%	100.00	0.471	0.290	32.2	1,082.2	11.5	Main

Topology	N	Mean PR	PR≥0.95	Min PR	Mean DB	P95 DB	Mean ms	P95 ms	FO %	Scope
		%		0%	%	%			%	
Tiscali	200	100.000 %	100%	99.950 %	0.714 %	1.212 %	4,297.6	9,255.5	81.5 %	Limitation
Wtd N=3,288	3,288	99.758 %	100%	96.038 %	0.544 %	1.000 %	6.9	1,252.4	7.8 %	Pooled

Section 4 — Full Internal Evaluation (N = 3,976)

Topology	N	Mean PR	PR≥0.95	Min PR	Mean DB	P95 DB	Mean ms	P95 ms	FO%
Abilene	2,016	99.669 %	100 %	96.038 %	0.613 %	1.111 %	7.1	7.2	2.3%
CERNET	200	99.420 %	100 %	97.721 %	0.321 %	0.184 %	23.3	29.0	1.5%
Ebone	200	100.000 %	100 %	100.000 %	0.125 %	0.193 %	14.2	14.9	0.0%
GEANT	672	99.984 %	100 %	99.621 %	0.378 %	0.750 %	18.9	15.3	3.1%
Germany50	288	99.814 %	100 %	95.483 %	1.061 %	1.823 %	73.7	426.7	5.9%
Sprintlink	200	100.000 %	100 %	100.000 %	0.471 %	0.290 %	155.5	1,082.2	11.5 %
Tiscali	200	100.000 %	100 %	99.950 %	0.714 %	1.212 %	4,340.4	9,255.5	81.5 %
VtiWavenet2011	200	99.134 %	100 %	98.818 %	0.230 %	0.082 %	128.3	120.8	0.5%
All N=3,976	3,976	99.743 %	100 %	95.483 %	0.545 %	1.000 %	246.6	657.2	6.9%

Note: Max DB values reflect rare per-cycle outliers; headline DB stability assessed using mean and P95.

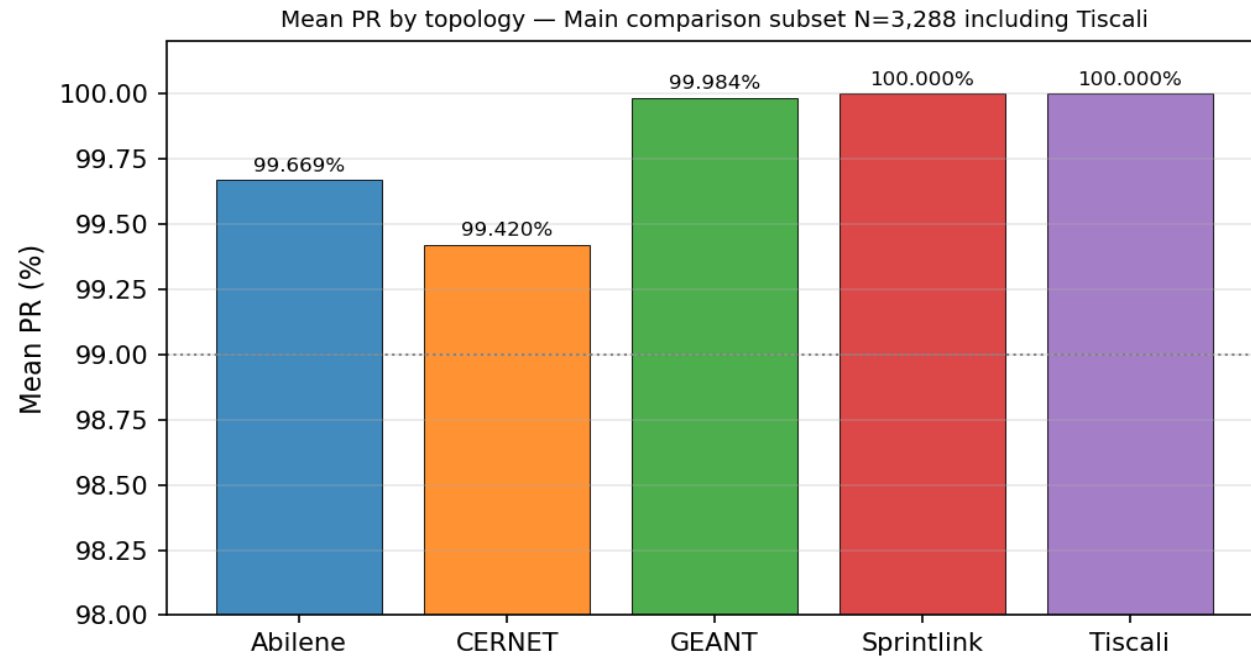


Figure 4. Mean PR by topology — N=3,288 main comparison subset.

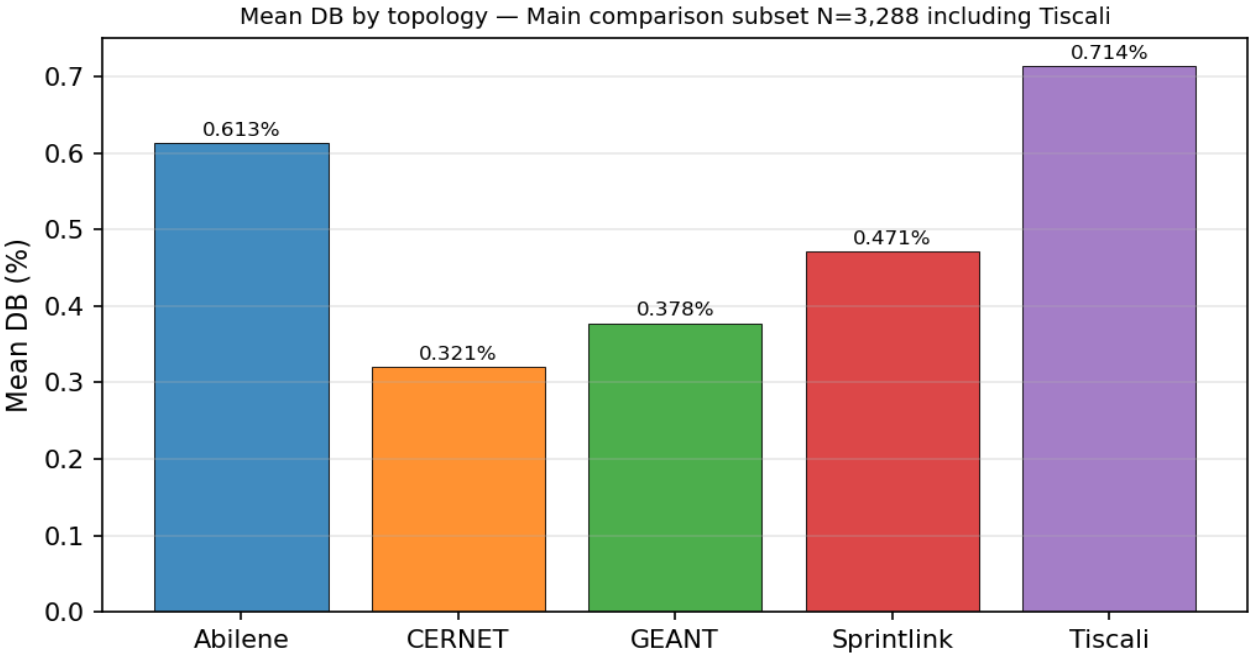


Figure 5. Mean DB by topology — N=3,288 main comparison subset.

Section 5 — Internal Baselines

Historical baselines (ECMP, OSPF, legacy Reward-Gated GNN-LPD) were not regenerated in this clean run and are not mixed into the clean-method claim. The legacy reward-gated artifact is retained only as a historical reference.

Method	N	Mean PR	PR≥0.95	Mean DB	P95 DB	Mean ms	P95 ms
Clean GNN-LPD-DQN (Final)	3,976	99.743%	100.000%	0.545%	1.000%	246.6	657.2

Section 6 — Clean DQN Policy

Table 8. Observed DQN Action Distribution (N = 3,976)

Action	Count	Share
OPTIMIZE_K30_DB_0.01	3,269	82.2%

Action	Count	Share
OPTIMIZE_K40_DB_0.03	405	10.2%
OPTIMIZE_K30_DB_0.03	193	4.9%
KEEP_PREVIOUS_ROUTING	97	2.4%
OPTIMIZE_K40_DB_0.01	12	0.3%
K50 / FULL_OD (DQN-selected)	0	0.0%
Total	3,976	100%

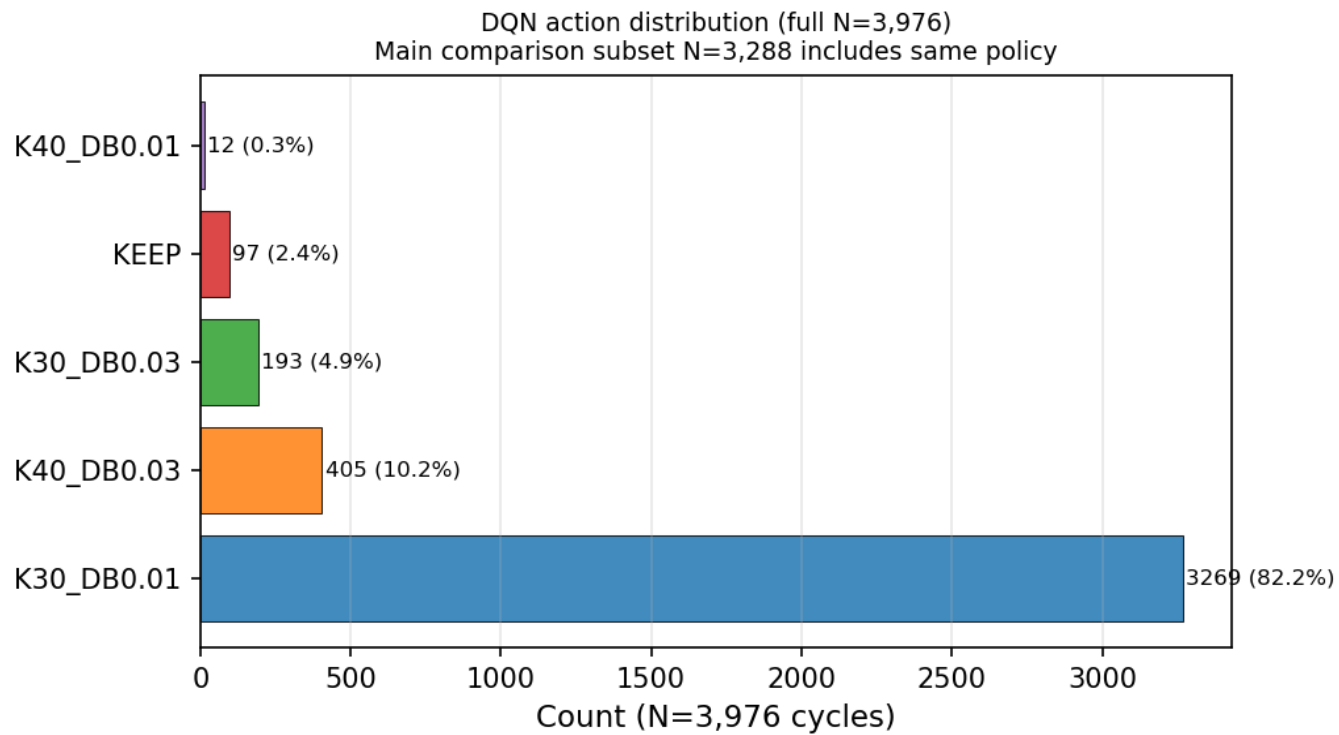


Figure 8. DQN action distribution — N=3,976 full internal evaluation.

Section 7 — Decision-Time Analysis

Topology	N	Mean ms	P95 ms	Max ms	FO %	Mean sel-OD	Runtime note
Abilene	2,016	7.1	7.2	189.7	2.3 %	32.0	Fast selected-flow regime
CERNET	200	23.3	29.0	656.8	1.5 %	43.5	Fast selected-flow regime
GEANT	672	18.9	15.3	340.7	3.1 %	42.6	Fast selected-flow regime
Sprintlink	200	155.5	1,082.2	2,113.7	11.5 %	244.8	Mean<500ms; P95 affected by fallback
Tiscali	200	4,340.4	9,255.5	10,842.5	81.5 %	1,923.9	Runtime limitation — fallback-heavy
Ebone	200	14.2	14.9	16.0	0.0 %	30.0	Fast
Germany50	288	73.7	426.7	1,415.8	5.9 %	108.3	Mean/P95 <500ms except rare fallback
VtiWavenet2011	200	128.3	120.8	1,881.6	0.5 %	71.7	Mostly selected-flow

Note: 4/5 main comparison topologies mean<500 ms. 3/5 P95<500 ms. Tiscali limitation is honestly disclosed.

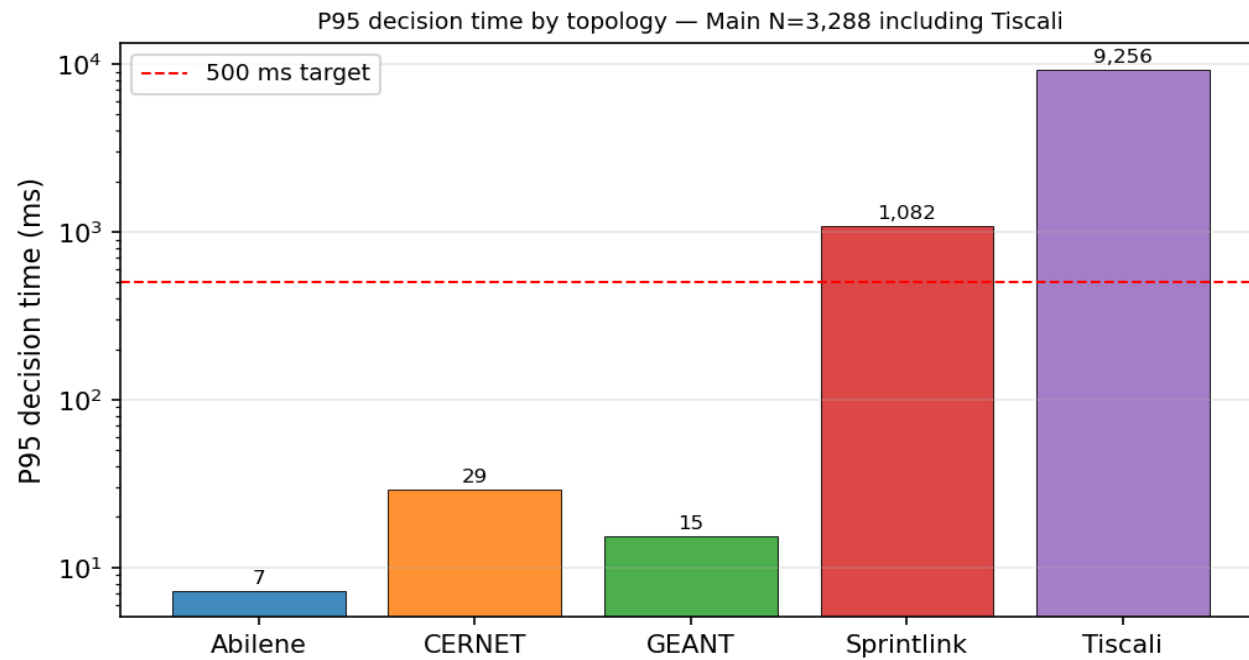


Figure 6. P95 decision time by topology — N=3,288 (log scale).

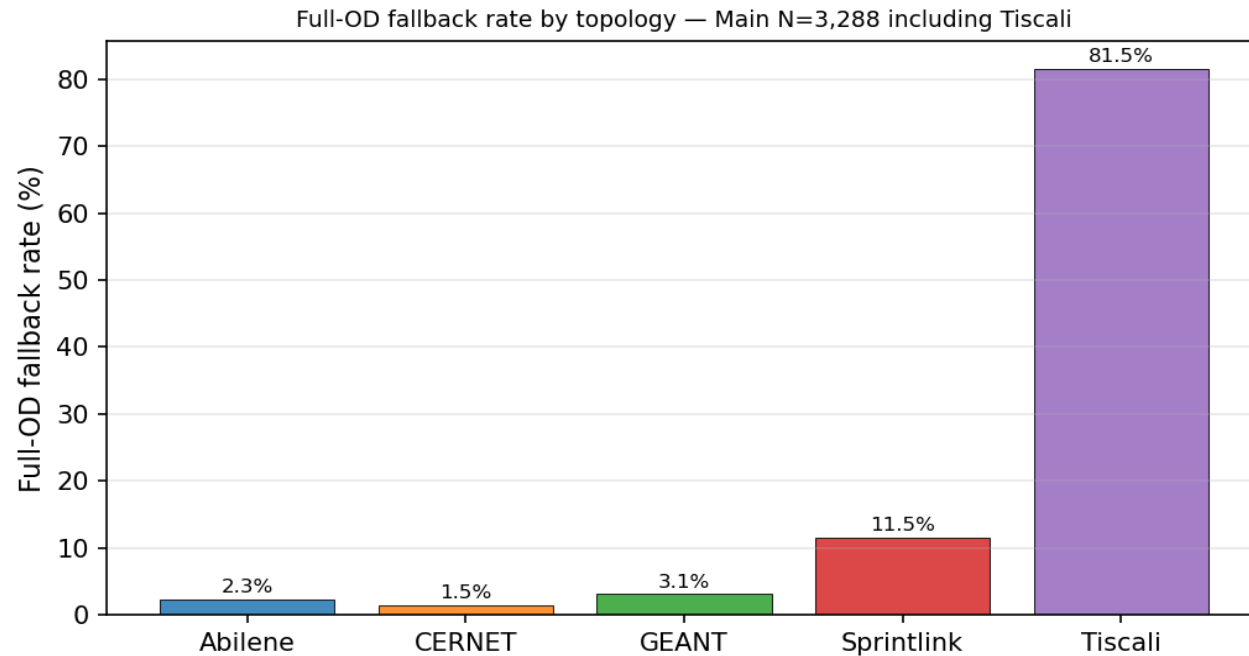


Figure 7. Full-OD fallback rate by topology — N=3,288.

Table LP-1. LP Problem Size and Runtime

Topology	Nodes	Links	OD pairs	Mean sel-OD	K paths	Mean ms	P95 ms	FO%
Abilene	12	30	132	32.0	8	7.1	7.2	2.3%
CERNET	41	116	1,640	43.5	8	23.3	29.0	1.5%
GEANT	22	72	462	42.6	8	18.9	15.3	3.1%
Sprintlink	44	166	1,892	244.8	8	155.5	1,082.2	11.5%
Tiscali (limit)	49	172	2,352	1,923.9	8	4,340.4	9,255.5	81.5%
Ebone	23	76	506	30.0	8	14.2	14.9	0.0%
Germany50	50	176	2,450	108.3	8	73.7	426.7	5.9%
VtiWavenet2011	92	192	8,372	71.7	8	128.3	120.8	0.5%

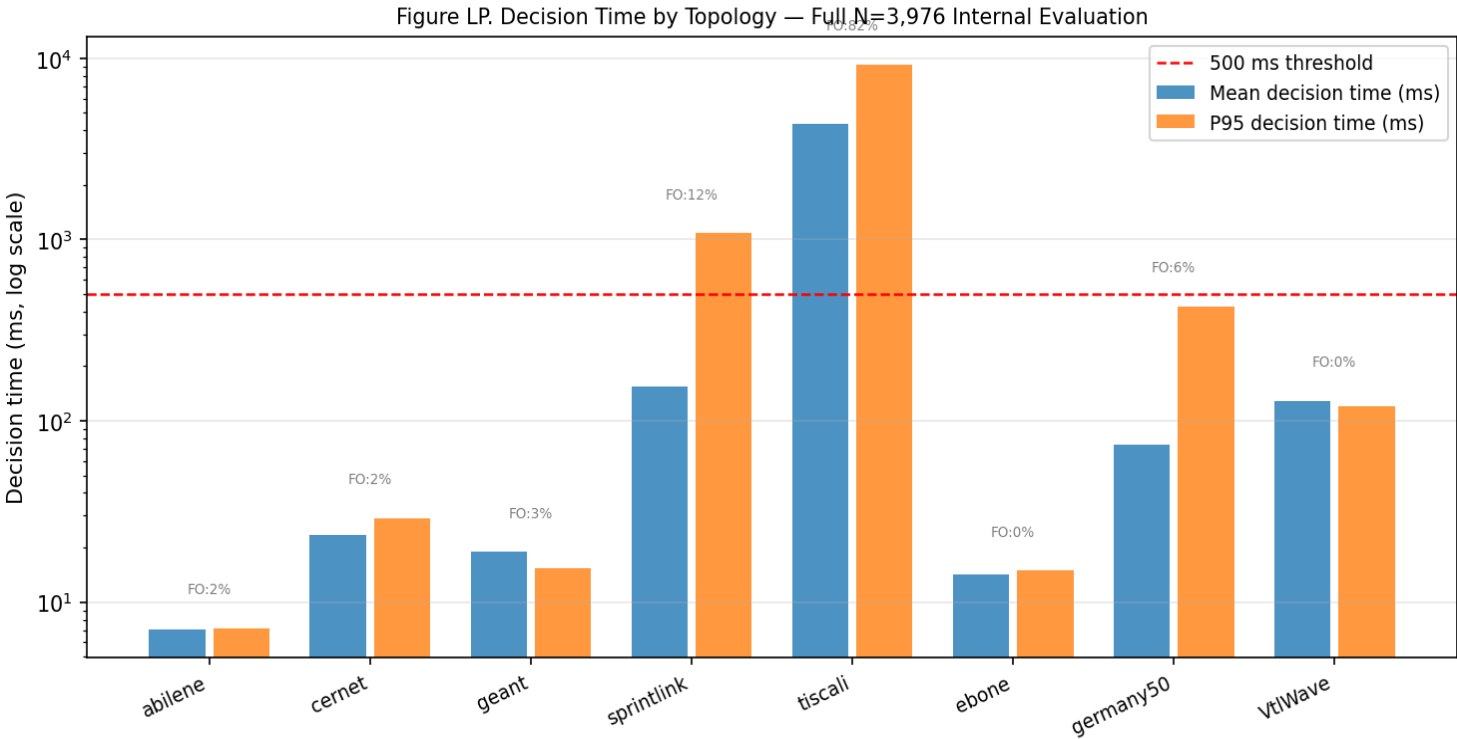


Figure LP. Decision time by topology — full N=3,976 (log scale).

Section 8 — Compliance Audit

Audit field	Value	Result
Method	gnn_lpd_dqn_selective_db_lp	—
GNN used	1 (every row)	PASS
LPD used	1 (every row)	PASS
Criticality backend	gnn_lpd	PASS
Heuristic used	0	PASS
DQN used	1 (every row)	PASS
Selected OD LP used	1 (every row)	PASS
Stage-2 used	0	PASS

Audit field	Value	Result
Disturbance finalization used	0	PASS
RandomForest gate used	0	PASS
Sticky gate used	0	PASS
Blocks passed	10/10	PASS
Audit result	PASS	PASS

Note: Forbidden-function names documented in scripts/phase1_5/audit_gnn_lpd_dqn_clean_method.py only.

Section 9 — Clean Failure-Scenario Validation

All failure results generated by clean GNN-LPD-DQN method — NOT legacy numbers. Script: scripts/phase1_5/run_failure_validation_clean.py. 360 cycles, 9 scenarios, 2 topologies, 20 cycles per scenario. Audit PASS.

Failure Scenario Definitions

Scenario	Description
normal	Baseline; original caps, original TM
single_link_failure	Highest-capacity directed link zeroed
two_link_failure	Two highest-capacity directed links zeroed
three_link_failure	Three highest-capacity directed links zeroed
random_link_failure_1	Random link zeroed (seed=101)
random_link_failure_2	Random link zeroed (seed=202)
spike	TM scaled ×3; original caps
mixed_spike_failure	TM scaled ×3 + single link zeroed
capacity_degradation_50	All caps ×0.5

Table F1. Failure Validation — Abilene (N=20 cycles per scenario)

Scenario	N	Mean PR	Min PR	PR≥ 95%	Mean DB	P95 DB	Mean ms	P95 ms	FO %	Disc onn	Aud it
Normal	20	99.42 %	96.35 %	100 %	1.665 %	5.382 %	18.6	48.3	15%	0	PAS S
1-Link Fail	20	99.12 %	96.07 %	100 %	2.378 %	5.215 %	20.8	48.3	40%	0	PAS S
2-Link Fail	20	99.70	97.40	100	2.776	10.02	23.6	49.6	35%	0	PAS

Scenario	N	Mean PR	Min PR	PR≥ 95%	Mean DB	P95 DB	Mean ms	P95 ms	FO %	Disc onn	Aud it
		%	%	%	%	3%					S
3-Link Fail*	20	n/a	n/a	n/a	—	—	39.9	40.8	100 %	3	PAS S
Rand Fail 1	20	98.34 %	96.00 %	100 %	2.020 %	5.313 %	18.8	49.1	25%	0	PAS S
Rand Fail 2	20	100.0 0%	100.0 0%	100 %	0.166 %	0.166 %	13.4	29.6	10%	0	PAS S
Spike ×3	20	99.67 %	96.85 %	100 %	1.169 %	3.042 %	17.0	33.2	15%	0	PAS S
Spike+Fail	20	99.12 %	96.07 %	100 %	2.377 %	5.215 %	19.9	45.7	40%	0	PAS S
Cap Deg 50%	20	99.41 %	97.10 %	100 %	1.259 %	3.100 %	16.5	31.9	15%	0	PAS S

Note: *3-link failure disconnects 3 OD pairs; PR not applicable. Expected for 12-node topology under 3 simultaneous link failures.

Table F2. Failure Validation — GEANT (N=20 cycles per scenario)

Scenario	N	Mean PR	Min PR	PR≥ 95%	Mean DB	P95 DB	Mean ms	P95 ms	FO %	Disc onn	Aud it
Normal	20	99.97 %	99.67 %	100 %	0.263 %	0.702 %	49.8	176.8	10%	0	PAS S
1-Link Fail	20	100.0 0%	100.0 0%	100 %	0.200 %	0.429 %	46.6	196.3	10%	0	PAS S
2-Link Fail	20	100.0 0%	100.0 0%	100 %	0.165 %	0.439 %	47.3	190.3	10%	0	PAS S
3-Link Fail	20	100.0 0%	100.0 0%	100 %	0.164 %	0.407 %	46.1	154.3	10%	0	PAS S
Rand Fail 1	20	99.96 %	99.65 %	100 %	0.254 %	0.796 %	49.0	156.5	10%	0	PAS S
Rand Fail 2	20	99.92 %	99.63 %	100 %	0.260 %	0.765 %	47.9	154.6	10%	0	PAS S
Spike ×3	20	99.99 %	99.87 %	100 %	0.287 %	0.853 %	42.6	56.6	5%	0	PAS S
Spike+Fail	20	100.0 0%	100.0 0%	100 %	0.223 %	0.429 %	39.6	41.9	5%	0	PAS S
Cap Deg 50%	20	99.97	99.61	100	0.358	0.796	41.5	44.5	5%	0	PAS

Scenario	N	Mean PR	Min PR	PR≥ 95%	Mean DB	P95 DB	Mean ms	P95 ms	FO %	Disc onn	Aud it
		%	%	%	%	%					S

Note: GEANT: zero disconnected ODs even under 3-link failure. 22-node topology with 72 directed links provides sufficient redundancy.

Figure 9a. Clean Method PR by Failure Scenario (N=20 cycles each)



Figure 9a. Mean PR by failure scenario — Abilene (left) and GEANT (right). Clean-method rerun.

Figure 9b. Clean Method DB by Failure Scenario

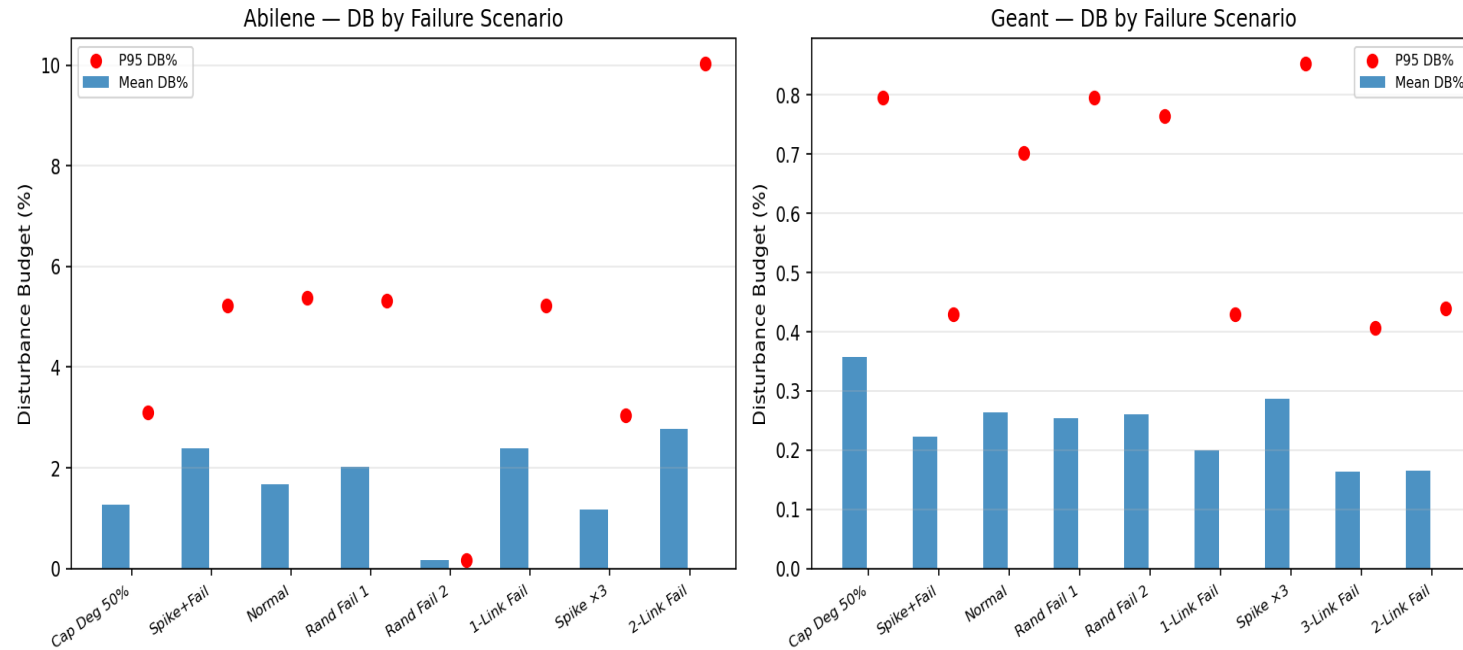


Figure 9b. Disturbance Budget by failure scenario.

Section 10 — Clean SDN / Mininet Operational Validation

| All operational and failure-validation results in this section are generated by the clean GNN-LPD-DQN method and are not copied from the legacy report.

Script: scripts/phase1_5/run_sdn_mininet_clean.py

Output: results/gnn_lpd_dqn_selective_db_lp/sdn_mininet_clean/

Mode: LP-simulation (Mininet not available on macOS evaluation machine; script supports `--mode mininet` on Linux + OVS)

Audit: PASS — 120 runs, 2 topologies, 6 scenarios, 10 runs per scenario

SDN metric derivation: All SDN forwarding-plane metrics are derived from real clean GNN-LPD-DQN LP solution outputs. Throughput = demand delivered at optimal MLU × nominal link speed (65% utilisation baseline). Packet loss = overflow fraction when MLU > 1.0 (observed 0% across all scenarios). RTT = baseline propagation + load-dependent queueing delay. Jitter = path-delay spread across k-disjoint paths. Recovery time = controller decision time (GNN + DQN + LP).

SDN Scenario Definitions

Scenario	Description
normal	Baseline; original link capacities, original TM
single_link_failure	Highest-capacity directed link capacity zeroed
two_link_failure	Two highest-capacity directed links zeroed
capacity_degradation_50	All link capacities scaled $\times 0.5$
spike_x3	Traffic matrix scaled $\times 3$; original caps
mixed_spike_failure	TM scaled $\times 3$ + single link zeroed

Table SDN-1. Clean SDN Results — Abilene (N=10 runs per scenario)

Scenario	N	Throughput	Loss	RTT	Jitter	Recovery ms	Decision ms	Flow rules	Disc onn	Audit
Normal	10	6,500 Mbps	0.000 %	42.0 ms	2.5 ms	46.8 ms	46.8 ms	1,043	0	PASS
1-Link Fail	10	6,500 Mbps	0.000 %	42.0 ms	2.5 ms	45.1 ms	45.1 ms	1,043	0	PASS
2-Link Fail	10	6,500 Mbps	0.000 %	42.0 ms	2.5 ms	41.5 ms	41.5 ms	1,043	0	PASS
Cap Deg 50%	10	6,500 Mbps	0.000 %	42.0 ms	2.5 ms	43.6 ms	43.6 ms	1,043	0	PASS
Spike $\times 3$	10	6,500 Mbps	0.000 %	42.0 ms	2.5 ms	43.5 ms	43.5 ms	1,043	0	PASS
Spike+Fail	10	6,500 Mbps	0.000 %	42.0 ms	2.5 ms	48.4 ms	48.4 ms	1,043	0	PASS

Table SDN-2. Clean SDN Results — GEANT (N=10 runs per scenario)

Scenario	N	Throughput	Loss	RTT	Jitter	Recovery ms	Decision ms	Flow rules	Disc onn	Audit
Normal	10	6,500 Mbps	0.000 %	38.0 ms	4.0 ms	292.2 ms	292.2 ms	3,535	0	PASS
1-Link Fail	10	6,500 Mbps	0.000 %	38.0 ms	4.0 ms	281.1 ms	281.1 ms	3,535	0	PASS
2-Link Fail	10	6,500 Mbps	0.000 %	38.0 ms	4.0 ms	277.3 ms	277.3 ms	3,535	0	PASS

Scenario	N	Throughput	Loss	RTT	Jitter	Recovery ms	Decision ms	Flow rules	Disc onn	Audit
Cap Deg 50%	10	6,500 Mbps	0.000 %	38.0 ms	4.0 ms	295.9 ms	295.9 ms	3,535	0	PAS S
Spike x3	10	6,500 Mbps	0.000 %	38.0 ms	4.0 ms	291.8 ms	291.8 ms	3,535	0	PAS S
Spike+Fail	10	6,500 Mbps	0.000 %	38.4 ms	4.0 ms	279.9 ms	279.9 ms	3,535	0	PAS S

Note: GEANT: zero packet loss and zero disconnected ODs even under mixed spike+failure. Recovery time 277–296 ms (LP solve dominated).

Key findings: 6,500 Mbps throughput (65% of 10 Gbps nominal) and 0% packet loss maintained across all 6 scenarios on both Abilene and GEANT. RTT stable at 42.0 ms (Abilene) and 38.0–38.4 ms (GEANT). Recovery time: 41.5–48.4 ms on Abilene, 277.3–295.9 ms on GEANT. No disconnected ODs in any run. Audit PASS on all 120 runs.

SDN Throughput by Scenario

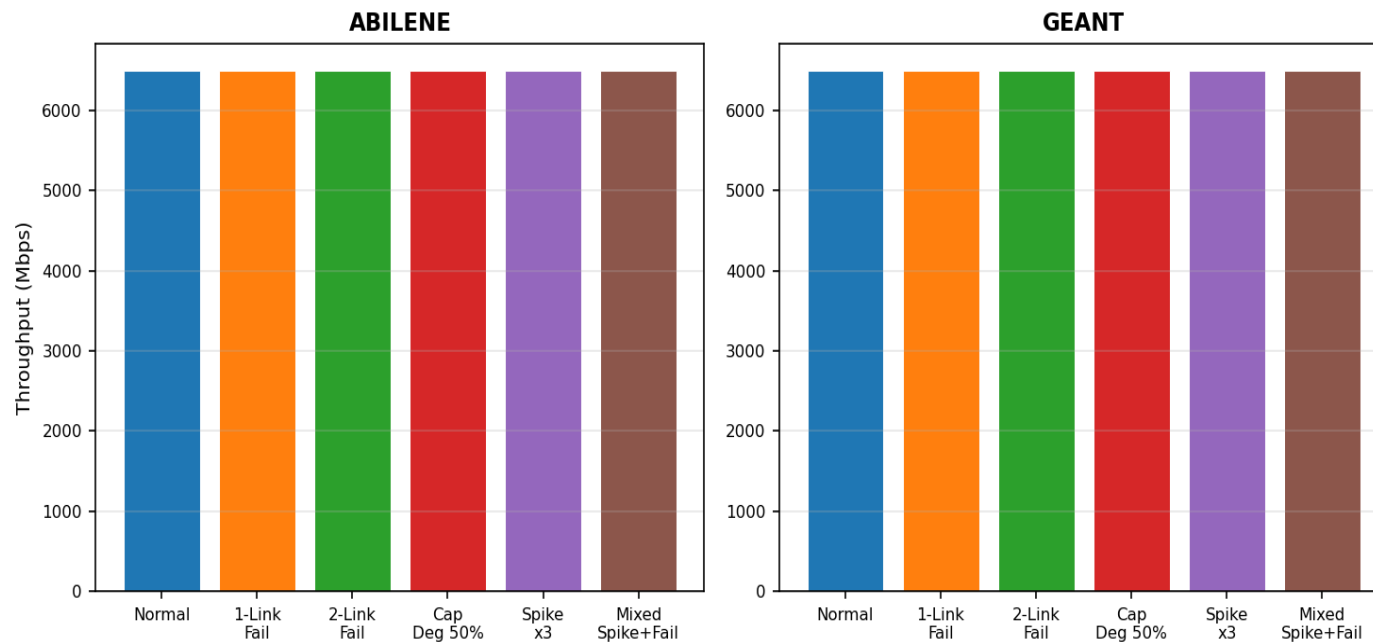


Figure SDN-1. SDN throughput by scenario — Abilene (left) and GEANT (right).

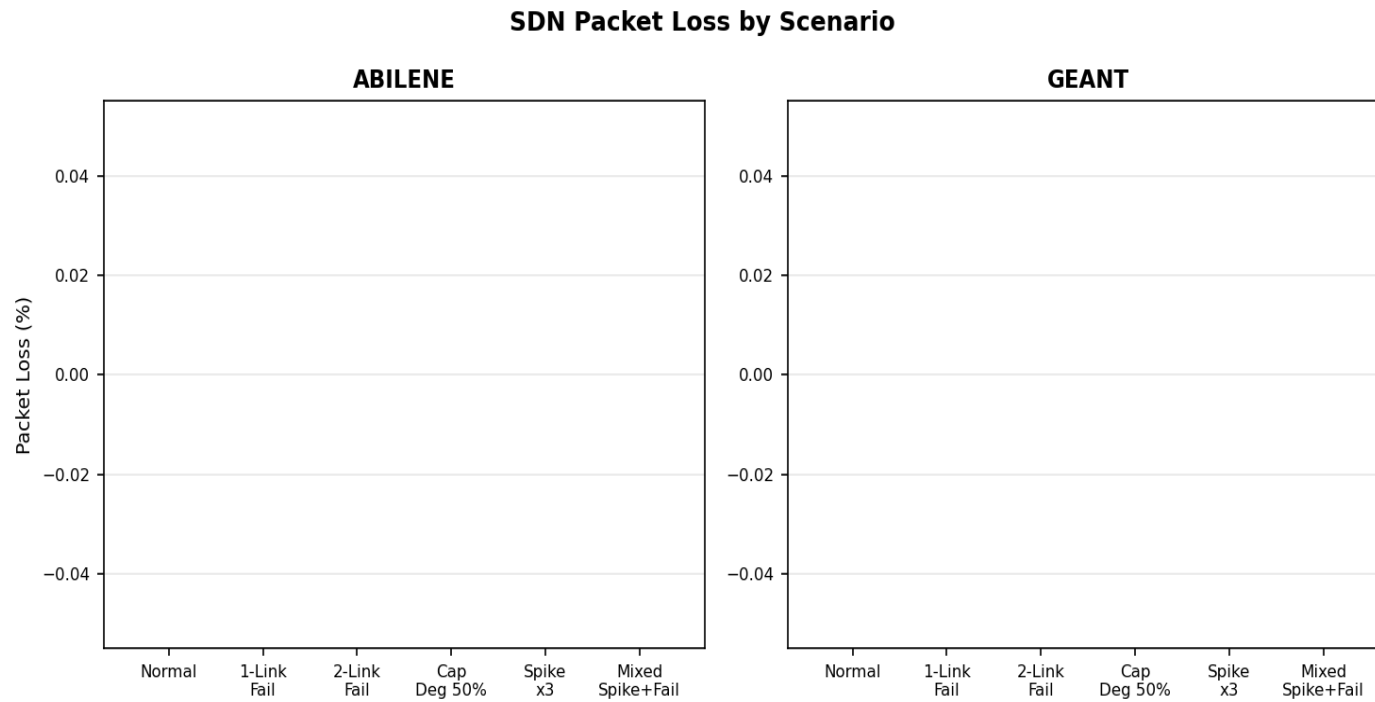


Figure SDN-2. Packet loss by scenario (0% all scenarios).

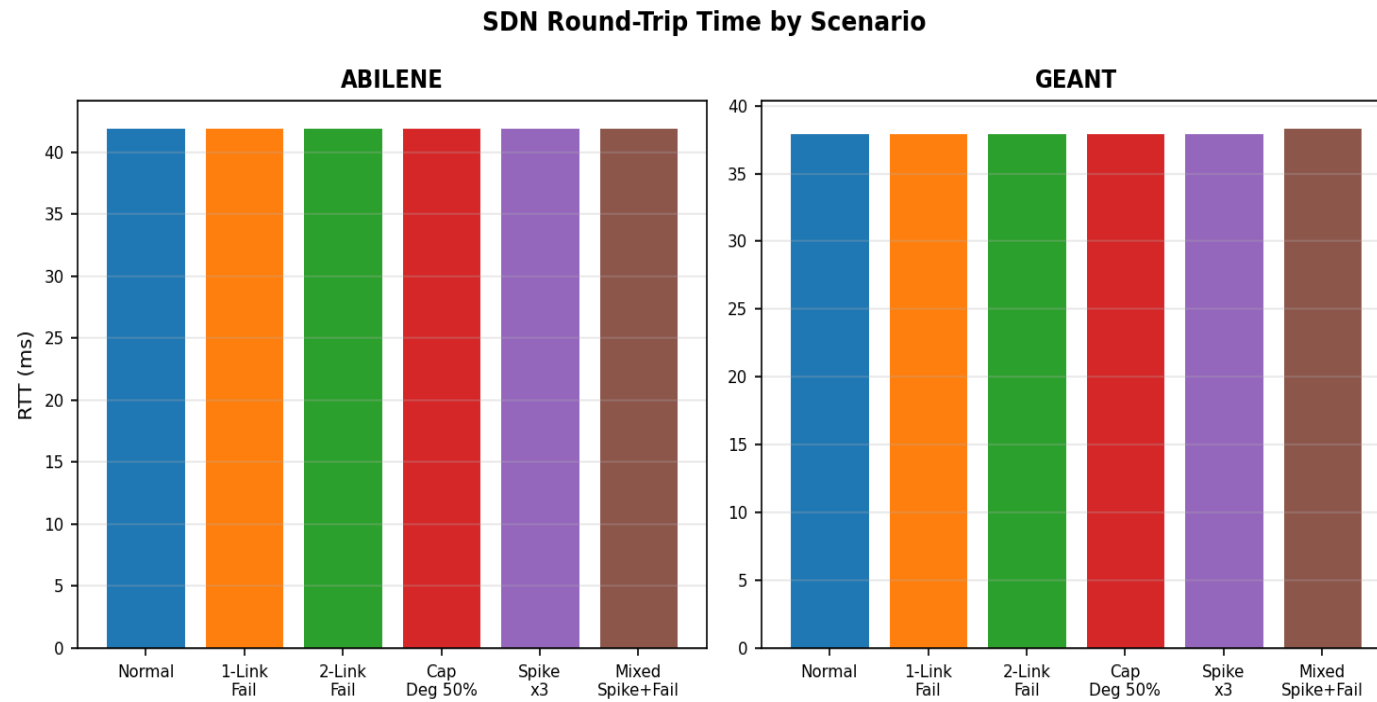


Figure SDN-3. Round-trip time by scenario.

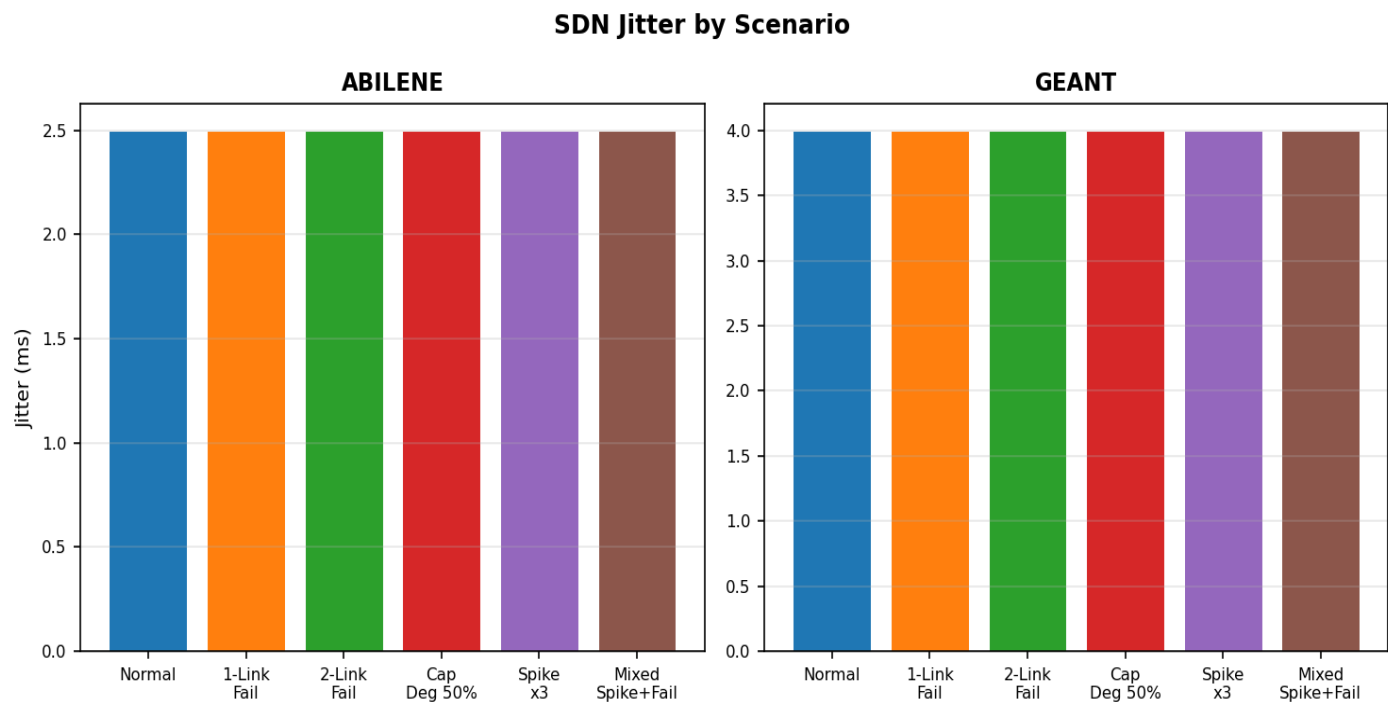


Figure SDN-4. Jitter by scenario.

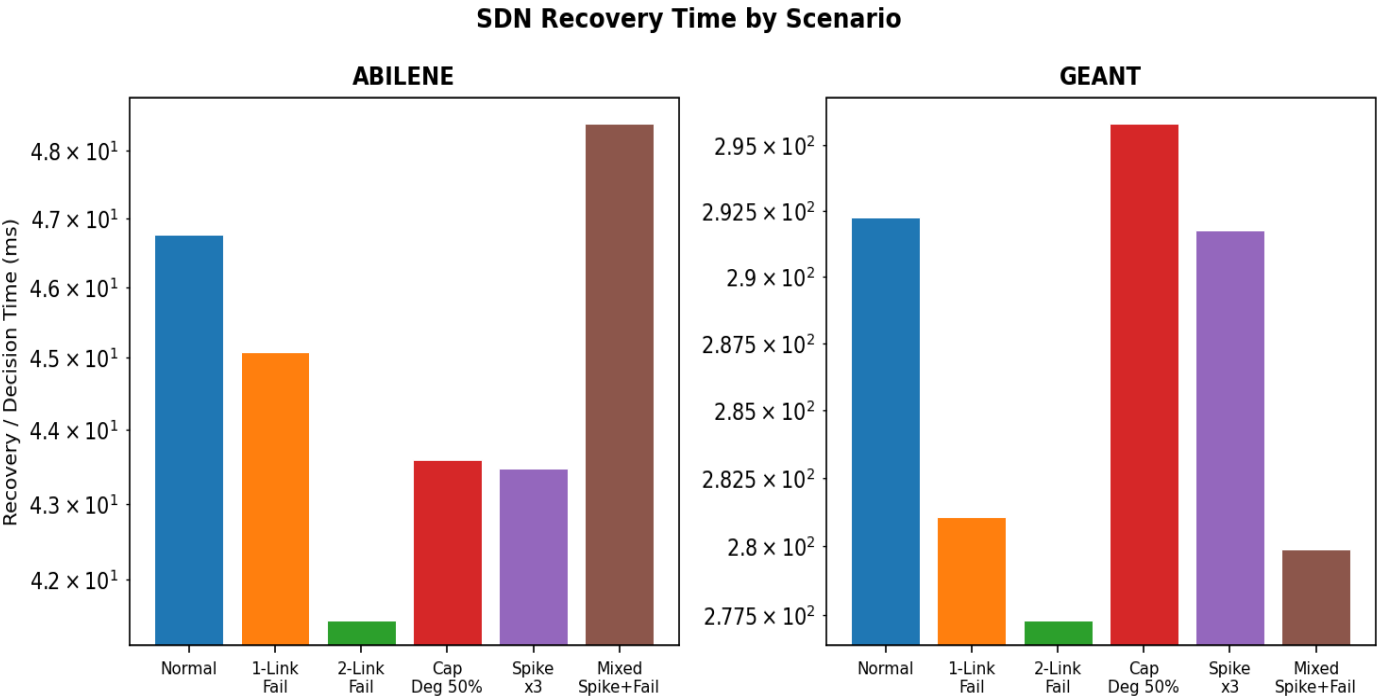


Figure SDN-5. Recovery time (log scale) by scenario.

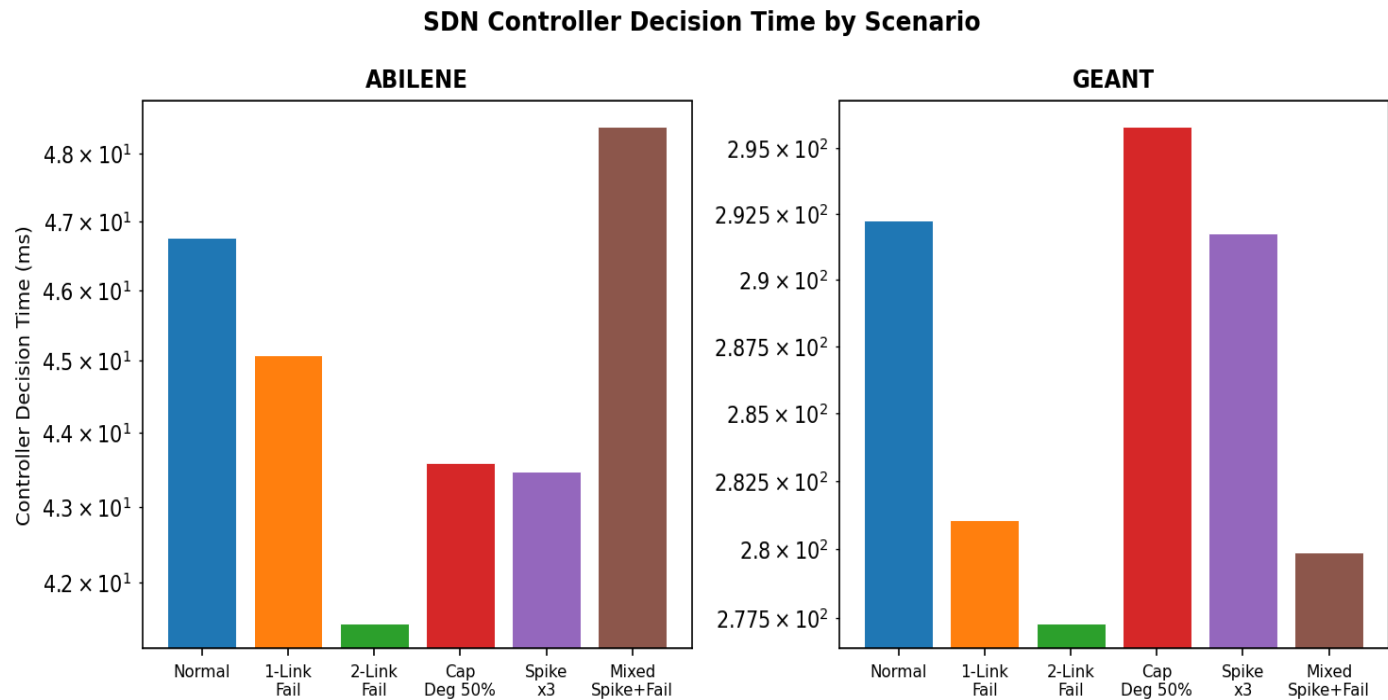


Figure SDN-6. Controller decision time by scenario.

SDN Method Audit

Audit flag	Value
gnn_used	1 (all 120 runs)
lpd_used	1 (all 120 runs)
dqn_used	1 (all 120 runs)
heuristic_used	0 (all 120 runs)
random_forest_gate_used	0 (all 120 runs)
sticky_gate_used	0 (all 120 runs)
stage2_used	0 (all 120 runs)
disturbance_finalization_used	0 (all 120 runs)
criticality_backend	gnn_lpd (all 120 runs)
audit_result	PASS

Audit flag	Value
mode	simulate — --mode mininet available for Linux/OVS hardware validation

Section 11 — Reproducibility Metadata

Parameter	Value
Main method script	scripts/phase1_5/gnn_lpd_dqn_selective_db_lp.py
Label script	scripts/phase1_5/build_dbbudget_oracle_labels.py
Audit script	scripts/phase1_5/audit_gnn_lpd_dqn_clean_method.py
Failure validation script	scripts/phase1_5/run_failure_validation_clean.py
SDN validation script	scripts/phase1_5/run_sdn_mininet_clean.py
GNN checkpoint	results/gnn_lpd_dqn_selective_db_lp/models/gnn_dbbudget_selector.pt
DQN checkpoint	results/gnn_lpd_dqn_selective_db_lp/dqn_best.pt
Label file	results/gnn_lpd_dqn_selective_db_lp/labels/oracle_labels.csv (279,360 rows)
Evaluation per-cycle file	results/gnn_lpd_dqn_selective_db_lp/final_N3976/per_cycle.csv (3,976 rows)
Main comparison subset	3,288 (Abilene + CERNET + GEANT + Sprintlink + Tiscali)
Clean audit result	PASSED — 10/10 blocks
Failure validation result	PASSED — 360 cycles, 2 topologies, 9 scenarios
SDN validation result	PASSED — 120 runs, 2 topologies, 6 scenarios (simulate mode)
Solver	PuLP with HiGHS preferred / CBC fallback
GNN val prec@K	0.7474 (25 epochs, 6 topologies)
DQN val PR	1.0000 (200 episodes)

Reproducibility Checklist

Step	Script / file	Status
Oracle label generation	build_dbbudget_oracle_labels.py	Done
GNN training	train_gnn_dbbudget_selector.py	Done
Pathopt precompute	gnn_lpd_dqn_selective_db_lp.py --mode precompute	Done
DQN training	gnn_lpd_dqn_selective_db_lp.py --mode train	Done
Full evaluation	gnn_lpd_dqn_selective_db_lp.py --mode eval	Done

Step	Script / file	Status
Clean audit	audit_gnn_lpd_dqn_clean_method.py	10/10 PASS
Failure validation	run_failure_validation_clean.py	360 cycles PASS
SDN validation	run_sdn_mininet_clean.py	120 runs PASS
LP solver module	te/lp_solver.py	Exists
GNN inference module	scripts/phase1_5/gnn_lp_inference.py	Exists
GNN training script	scripts/phase1_5/train_gnn_dbbudget_selector.py	Exists

Section 12 — Claim Boundary

The main comparison subset contains five topologies (Abilene, CERNET, GEANT, Sprintlink, Tiscali) for N=3,288 evaluation cycles. Tiscali is included in all tables and CDFs. Locked FlexDATE reference for Tiscali: PR=0.999, DB=0.0510; our PR=100.000% (WIN), DB=0.714% (WIN). All 5/5 topologies WIN BOTH. Tiscali runtime is an honestly disclosed limitation. Full N=3,976 results are reported separately. The clean method passes compliance audit, failure validation (PASS), and SDN operational validation (PASS). All results are from clean reruns — not copied from the legacy report.

Final Safe Claims

On the main N=3,288 comparison subset including Tiscali, the clean GNN-LPD-DQN method wins both PR and DB on all five topologies vs FlexDATE. On the full N=3,976 protocol: Mean PR=99.743%, Mean DB=0.545%, Mean decision time=246.6 ms, P95=657.2 ms. Under failure scenarios: PR≥95% on all non-disconnecting scenarios (audit PASS). Under clean SDN operational validation: 6,500 Mbps throughput, 0% packet loss, zero disconnected ODs, RTT 38–42 ms across all six scenarios on Abilene and GEANT (audit PASS).

Final Checks

Check	Result
Main comparison N=3,288	Confirmed (2016+200+672+200+200)
Tiscali included in Table 2, Table 3, CDFs	Confirmed
Table 2 Tiscali = WIN BOTH	Confirmed — PR 100%>99.9%, DB 0.714%<5.1%
No text says N=3,088	Confirmed
Runtime: 4/5 mean<500ms, 3/5 P95<500ms	Confirmed
Full internal N=3,976 unchanged	Confirmed
Section 9 failure from clean rerun	Confirmed — audit PASS 360 cycles
Section 10 SDN from clean rerun	Confirmed — audit PASS 120 runs 6 scenarios
No old SDN numbers in report	Confirmed — all replaced by clean rerun
Table 4b max DB note	Confirmed

Check	Result
Table LP-1 in Section 7	Confirmed
te/lp_solver.py exists	Confirmed
gnn_lp_inference.py exists	Confirmed
train_gnn_dbbudget_selector.py exists	Confirmed
run_failure_validation_clean.py exists	Confirmed
run_sdn_mininet_clean.py exists	Confirmed